Alessandro **Sanvito**

Software and AI Engineer

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Experience

Optiver — *Amsterdam, Netherlands*

Software Engineer — *August 2023 - Current*

- Expanded the desk beyond index options: built baseline systematic strategies for new product groups from scratch, enabling researchers to rapidly optimize and generate new alpha.
- Selected candidate signals and defined the data-quality and validation criteria each baseline must meet before handoff.
- Built data pipelines processing 10+ TB of instrument-level data with Spark, Polars, and DuckDB, cutting core backtest runtime by 60%; moved ML model training to AWS.

Mercedes-Benz (AI-SEE Project) — *Stuttgart, Germany*

AI Research Intern and Master Thesis Student — *May 2022 - June 2023*

- Researched NeRFs and diffusion models for 3D scene reconstruction and human avatar modeling from monocular video in autonomous driving systems.
- Outperformed state-of-the-art baselines in PSNR and SSIM when reconstructing scenes in fog, haze, and limited viewpoints; published first-author at IEEE IV 2024 and co-authored at ICCV 2023.

Education

KTH Royal Institute of Technology — *Stockholm, Sweden*

MSc. in ICT Innovation, Data Science — *Sept. 2020 - May 2023*

- Final grade: A - Excellent
- Implemented from scratch a diverse range of neural network architectures in NumPy, including MLPs, Hopfield Networks, and Deep Belief Networks.

Polytechnic University of Milan — *Milan, Italy*

MSc. in Computer Science and Engineering — *Sept. 2020 - Jul. 2023*

- **1st place (academic)** at the international ACM RecSys Challenge 2021: co-developed the winning ML solution on Twitter's 1B-interaction dataset with the university team under prof. Paolo Cremonesi, using Dask, CatBoost, and XGBoost.
- Final grade: 110/110 cum laude

Polytechnic University of Milan — *Milan, Italy*

BSc. in Engineering of Computing Systems — *Sept. 2017 - Jul. 2020*

- Final grade: 109/110

Publications

2024	**HINT: Learning Complete Human Neural Representations from Limited Viewpoints**, **Sanvito Alessandro** (first author), et al.	*IEEE IV 2024*
2023	**ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Renderings**, Ramazzina Andrea, et al., incl. **Sanvito Alessandro**	*ICCV 2023*
2022	**United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources**, Maldini Pietro, **Sanvito Alessandro**, Surricchio Mattia	*ACM recsys'22*
2021	**Lightweight and Scalable Model for Tweet Engagements Predictions in a Resource-constrained Environment**, Carminati Luca, et al., incl. **Sanvito Alessandro**	*ACM recsys'21*

Skills

Programming	Python, SQL, Java, C++, C
Research & ML	PyTorch, generative models (NeRFs, Diffusion), computer vision, recommender systems, gradient boosting (CatBoost, XGBoost), scikit-learn
Data Engineering	Spark, Polars, DuckDB, PostgreSQL, Delta Lake, Pandas, NumPy, MLlib, AWS
Tools & Platforms	Git, Docker, Linux, CI/CD, pytest